Sharp L_p–L_2 moment comparison for centred log-concave random variables, with equality cases

Anonymous

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Abstract

For a nondegenerate mean-zero log-concave random variable X, we determine the sharp constants in $\|X\|_p \geq m_p\|X\|_2$ $(1 < p < 2)$ and $\|X\|_p \leq M_p\|X\|_2$ $(p > 2)$. The extremiser changes from the symmetric Laplace law to a centred exponential law at $p^* = 1.6824572\ldots$, the unique solution in $(1, 2)$ of $\Gamma(p+1)^{1/p}/\sqrt{2} = (\mathbb{E}|E-1|^p)^{1/p}$, with E standard exponential. This proves the variance-normalised moment conjecture of Melbourne, Roysdon, Tang and Tkocz. Away from p^* and 2, the extremiser is unique up to nonzero scaling; at p^* there are exactly the two stated families. At $p = 2$ every admissible law is extremal. We give a self-contained, equality-sensitive proof of Eitan's real-exponent reduction using crossing arguments and an explicit Chebyshev system. The remaining two-variable optimisation is proved analytically for $p \geq 7$ and by rigorous ball-arithmetic enclosures for $1 < p < 7$, with separate derivative estimates at the degenerate boundaries.

1 Introduction

For a random variable X write $\|X\|_p = (\mathbb{E}|X|^p)^{1/p}$. Let $\mathcal{L}$ be the class of nondegenerate mean-zero log-concave random variables, that is, those with a density e^{-V} for a proper convex $V : \mathbb{R} \to (-\infty, +\infty]$ (so supports may be half-lines or intervals; $E - 1 \in \mathcal{L}$). Melbourne, Roysdon, Tang and Tkocz [7] determined the sharp constants comparing $\|X\|_p$ with $\|X\|_1$ on $\mathcal{L}$, finding a phase transition in the extremiser. In their concluding remarks [7, Sec. 3.1] they ask for the sharp constants in the variance-normalised problem,

$$m_p = \inf_{X\in\mathcal{L}} \frac{\|X\|_p}{\|X\|_2} \quad (1 < p < 2), \qquad M_p = \sup_{X\in\mathcal{L}} \frac{\|X\|_p}{\|X\|_2} \quad (p > 2),$$

and, based on numerical experiments, conjecture that the extremiser is the symmetric exponential for $1 < p < p^* \approx 1.68$ and the one-sided exponential for $p > p^*$. The phase transition and the value $p^* \approx 1.68$ were thus found numerically by Melbourne, Roysdon, Tang and Tkocz; our contribution is a proof. Eitan [3, Theorem 1.4] (Theorem 4 in the arXiv version) proved the case of even integers $p > 2$.

At $p = 0$ we use the limiting convention $\|X\|_0 = \exp(\mathbb{E}\log|X|)$. The logarithm is integrable for the laws considered here: the density is bounded near zero and has exponential tails. The new results below concern $p > 1$.

Let E, E' be independent standard exponential random variables, $L = E - E'$ (symmetric exponential) and $F(p) = \mathbb{E}|E-1|^p$. Then $\|L\|_p/\|L\|_2 = \Gamma(p+1)^{1/p}/\sqrt{2}$ and $\|E-1\|_p/\|E-1\|_2 = F(p)^{1/p}$.

Theorem 1. *There is a unique $p^* \in (1,2)$ with $\Gamma(p^*+1)^{1/p^*}/\sqrt{2} = F(p^*)^{1/p^*}$; numerically $p^* = 1.6824572084\ldots$. Moreover:*

- *for $1 < p \le p^*$, $m_p = \Gamma(p+1)^{1/p}/\sqrt{2}$, attained by the symmetric exponential;*
- *for $p^* \le p < 2$, $m_p = F(p)^{1/p}$, attained by the one-sided exponential $E-1$;*
- *for $p > 2$, $M_p = F(p)^{1/p}$, attained by $E-1$.*

With [7, Theorem 1], which treats $-1 < p \le 1$ (symmetric exponential extremal), Theorem 1 gives the sharp constants comparing L_p with L_2 on $\mathcal{L}$ for all $p > -1$; for even $p > 2$ it recovers [3, Theorem 1.4]. We also classify the cases of equality. Here $X \overset{d}{=} Y$ means equality in law.

Theorem 2 (Equality cases)**.** *Let $X \in \mathcal{L}$, and let p^*, m_p, M_p be as in Theorem 1.*

(i) If $1 < p < p^*$, then $\|X\|_p = m_p\|X\|_2$ if and only if $X \overset{d}{=} \lambda(E-E')$ for some $\lambda > 0$.

(ii) If $p = p^*$, then $\|X\|_p = m_p\|X\|_2$ if and only if $X \overset{d}{=} \lambda(E-E')$ or $X \overset{d}{=} \pm\lambda(E-1)$ for some $\lambda > 0$.

(iii) If $p^* < p < 2$, then $\|X\|_p = m_p\|X\|_2$ if and only if $X \overset{d}{=} \pm\lambda(E-1)$ for some $\lambda > 0$.

(iv) If $p > 2$, then $\|X\|_p = M_p\|X\|_2$ if and only if $X \overset{d}{=} \pm\lambda(E-1)$ for some $\lambda > 0$.

Contribution and proof architecture. The new optimisation and complete classification in Theorems 1–2 concern the remaining real exponents. Eitan's reduction already covers real $1 < a < b$; neither that scope nor the crossing method is claimed as new. Section 2 supplies a self-contained, equality-sensitive proof of the reduction, following Eitan [3, Sections 2–3] and the crossing methods of Eskenazis, Nayar and Tkocz [4]. Sections 4–8 establish the reduced optimisation, and Section 9 extracts the strictness needed for the equality classification. All arguments and proof-critical computational material are included in this paper and its accompanying public package; no earlier draft is required.

Related work. Murawski [9] proved the mean-zero bound $\|X\|_p \le \frac{p}{q}\|X\|_q$ $(p \ge q \ge 1)$ and sharp shifted-exponential results for general (not necessarily centred) log-concave variables. Eitan conjectured [3, Conjecture 16 in arXiv v1] that for even $p < q$ the extremal Z_s is $s \in \{0,1\}$; Theorem 1 shows that this endpoint behaviour fails for real exponents, since for $(p,2)$ with $p < p^*$ the extremal parameter is $s = \frac{1}{2}$. Theorem 3 reduces that conjecture, with its equality case, to an inequality within the family Z_s; we do not address it. Since $Z_s/\|Z_s\|_2$ is a two-term centred sum of exponentials, the problem also belongs to the study of sharp moment bounds for sums of exponential random variables [2, 1].

Method. By Theorem 3 (Section 2) the extremal ratio is attained within the family $Z_s = s(E-1) - (1-s)(E'-1)$, $0 \le s \le 1$, and only there, so the theorem is equivalent to a two-parameter inequality for $G(s,p) = \frac{1}{p}\log\mathbb{E}|Z_s|^p - \frac{1}{2}\log\mathbb{E}Z_s^2$ (Section 3). The inequality is degenerate along $s = 0$ and $s = 1/2$ (where $\partial_s G$ vanishes), along $p = 2$ (where $G \equiv 0$), and at the crossing $p = p^*$. We prove it analytically for $p \ge 7$ (Section 4). For $1 < p < 7$ we partition $[0, \frac{1}{2}] \times (1,7]$ into regions with explicit margins, certified by ball arithmetic, and regions around the degenerate lines, where second-order lemmas reduce the claim to certified sign conditions on derivatives (Sections 5–7). All the certified inequalities are strict, and this gives the equality cases (Section 9).

A concrete sharp bound. At $p = 4$, direct integration gives $F(4) = \mathbb{E}(E-1)^4 = 24 - 24 + 12 - 4 + 1 = 9$. Thus every $X \in \mathcal{L}$ satisfies $\mathbb{E}X^4 \le 9(\mathbb{E}X^2)^2$, with equality exactly for a scaled or reflected centred exponential. This even-integer case was already known from Eitan; it illustrates the normalisation rather than a new fourth-moment inequality. In contrast, at $p = 3/2 < p^*$ the sharp lower ratio is $\Gamma(5/2)^{2/3}/\sqrt{2}$ and only the Laplace family attains it. At $p = 2$ the ratio equals one for every admissible law. Section 10 also shows that a nearly extremal ratio at a fixed $p > 1$, $p \neq 2$, forces closeness in total variation and every finite Wasserstein distance to the corresponding variance-normalised equality set. This qualitative consequence has no explicit deficit-to-distance rate.

2 A self-contained reduction (after Eitan)

Throughout, E, E' are independent standard exponential random variables and

$$Z_s = s(E-1) - (1-s)(E'-1), \qquad 0 \le s \le 1.$$

Thus $Z_{1/2} = \frac{1}{2}(E - E')$, $Z_1 = E - 1$, $Z_0 = 1 - E' \overset{d}{=} -(E-1)$, and $Z_{1-s} \overset{d}{=} -Z_s$ (exchange E and E').

Relation to Eitan's work. The two theorems below give self-contained versions of Theorems 12 and 14 of Eitan's arXiv:2110.14382v1 [3]; Theorem 12 corresponds to Theorem 3.1 of the published article. The real-exponent reduction is already established there. Although its arXiv Theorem 12 does not state equality explicitly, Theorem 14 does, and the proof of Theorem 4 uses an equality characterisation for the reduction. Thus we do not claim priority for either the reduction or equality-sensitive reasoning. Our detailed comparison uses the arXiv text; the published full text was not inspected. The proofs below follow Eitan's crossing method and its antecedents [4], using Rolle's theorem, determinants, convexity, dominated convergence and the intermediate value theorem. Their role is to make the logical route to the complete extremiser classification explicit.

Theorem 3 (Reduction with equality)**.** *Let $1 < a < b$ be real and $X \in \mathcal{L}$. There exist $s_0 \in [0,1]$ and $\lambda > 0$ such that $\mathbb{E}|X|^a = \mathbb{E}|\lambda Z_{s_0}|^a$ and*

$$\mathbb{E}|X|^b \le \mathbb{E}|\lambda Z_{s_0}|^b,$$

with equality if and only if $X \overset{d}{=} \lambda Z_{s_0}$. Consequently the maximum below exists and

$$\frac{\|X\|_b}{\|X\|_a} \le \max_{0 \le s \le 1} \frac{\|Z_s\|_b}{\|Z_s\|_a},$$

with equality if and only if $X \overset{d}{=} \lambda Z_{s_1}$ for some $\lambda > 0$ and some s_1 at which the maximum is attained.

Theorem 4 (Odd moments)**.** *Let $1 < a < r$ be real and $X \in \mathcal{L}$. Then*

$$\frac{\mathbb{E}[|X|^r \operatorname{sgn} X]}{\|X\|_a^r} \le \frac{\mathbb{E}[|E-1|^r \operatorname{sgn}(E-1)]}{\|E-1\|_a^r},$$

with equality if and only if $X \overset{d}{=} \lambda(E-1)$ for some $\lambda > 0$.

2.1 Tools

Lemma A (log-concave preliminaries)**.** *Let $X \in \mathcal{L}$ have density e^{-V}, where $V : \mathbb{R} \to (-\infty, \infty]$ is convex. Then $I := \{V < \infty\}$ is an interval with nonempty interior; V is finite and convex on $\operatorname{int} I$, and $f := e^{-V}\mathbf{1}_{\operatorname{int} I}$ is a version of the density; and there are $C, \kappa > 0$ with $f(x) \le Ce^{-\kappa|x|}$ for all x. In particular $\mathbb{E}|X|^t < \infty$ for every $t \ge 0$.*

Proof. I is convex and has positive Lebesgue measure because $\int e^{-V} = 1$, so it is an interval with nonempty interior; its boundary has at most two points, a null set. Fix $x_0 \in \operatorname{int} I$ and $m \in \partial V(x_0)$ (the subdifferential is nonempty at interior points). Then $V(x) \ge V(x_0) + m(x - x_0)$ for all $x \in \mathbb{R}$, so $f(x) \le e^{-V(x_0)}e^{-m(x-x_0)}$ everywhere; in particular f is bounded on every bounded set, and on a bounded side of I it vanishes beyond the endpoint.

Suppose I is unbounded above. If $V(x) \le V(x_0)$ for all $x > x_0$, then $e^{-V} \ge e^{-V(x_0)}$ on (x_0, ∞), contradicting integrability. So there is $x_1 > x_0$ with $V(x_1) > V(x_0)$, and by convexity, for $x \ge x_1$,

$$V(x) \ge V(x_1) + \kappa_+(x - x_1), \qquad \kappa_+ = \frac{V(x_1) - V(x_0)}{x_1 - x_0} > 0.$$

The same argument applies below if I is unbounded below. Hence on each unbounded side f decays exponentially beyond some point; on each bounded side f vanishes beyond the endpoint; and in between f is bounded by the affine-minorant bound. Combining these gives $C, \kappa > 0$ with $f(x) \le Ce^{-\kappa|x|}$. □

Lemma B (Descartes' rule for generalised polynomials)**.** *Let $\lambda_0 < \cdots < \lambda_n$ be real and $c = (c_0, \dots, c_n) \ne 0$. Then $P(t) = \sum_i c_i t^{\lambda_i}$ has at most $\mathrm{V}(c)$ distinct zeros in $(0, \infty)$, where $\mathrm{V}(c)$ is the number of sign changes in c after zero entries are deleted.*

Proof. Induction on $\mathrm{V}(c)$. If $\mathrm{V}(c) = 0$, all nonzero terms $c_i t^{\lambda_i}$ have the same sign for $t > 0$, and at least one is nonzero, so $P(t) \ne 0$. If $\mathrm{V}(c) \ge 1$, choose consecutive nonzero entries $c_j, c_{j'}$ $(j < j')$ of opposite signs and $\mu \in (\lambda_j, \lambda_{j'})$. The function $\psi(t) = t^{-\mu}P(t)$ has the same zeros as P on $(0, \infty)$, and

$$\psi'(t) = \sum_i c_i(\lambda_i - \mu)t^{\lambda_i - \mu - 1}.$$

Its exponents are distinct, and its coefficients are nonzero exactly where $c_i \ne 0$. Passing from c to these coefficients flips the sign of every entry with $i \le j$ and keeps every entry with $i \ge j'$; this preserves the sign changes inside each group and removes the one between c_j and $c_{j'}$. So the coefficients of ψ' have $\mathrm{V}(c) - 1$ sign changes. If P had N distinct zeros, Rolle's theorem would give ψ' at least $N - 1$ distinct zeros, and by induction $N - 1 \le \mathrm{V}(c) - 1$. □

This is the classical rule of Laguerre [8, Part V, Ch. 1, Sec. 6]; the proof is included for completeness.

Lemma C. *For $c \in \mathbb{R}^{n+1} \setminus \{0\}$ let $c_i^* = (-1)^i c_i$. Then $\mathrm{V}(c) + \mathrm{V}(c^*) \le n - \mathbf{1}[c_0 = 0]$.*

Proof. Consider consecutive nonzero entries at indices $i < i'$, with gap $g = i' - i \ge 1$. This pair contributes $\mathbf{1}[c_i c_{i'} < 0] + \mathbf{1}[(-1)^g c_i c_{i'} < 0]$ to the left side. If g is odd the contribution is exactly $1 \le g$; if g is even it is 0 or $2 \le g$. Summing over pairs gives at most $i_{\text{last}} - i_{\text{first}} \le n - \mathbf{1}[c_0 = 0]$. □

Proposition D (Chebyshev property)**.** *Let $1 < a < r < b$ and set*

$$u_0 = 1, \quad u_1 = x, \quad u_2 = |x|^a, \quad u_3 = |x|^r \operatorname{sgn} x, \quad u_4 = |x|^b.$$

These are continuous on $\mathbb{R}$. For each $n \in \{0, \dots, 4\}$, every nontrivial combination $\varphi = \sum_{i \le n} c_i u_i$ has at most n distinct zeros in $\mathbb{R}$.

Proof. Let $(\lambda_0, \dots, \lambda_4) = (0, 1, a, r, b)$. For $t > 0$, $u_i(t) = t^{\lambda_i}$ and $u_i(-t) = (-1)^i t^{\lambda_i}$. So on $(0, \infty)$, $\varphi = \sum c_i t^{\lambda_i}$ has at most $\mathrm{V}(c)$ distinct zeros by Lemma B; $t \mapsto \varphi(-t) = \sum c_i^* t^{\lambda_i}$ has at most $\mathrm{V}(c^*)$ distinct zeros in $(0, \infty)$; and 0 is a zero if and only if $c_0 = 0$. Lemma C bounds the total by n. □

The same proof applies to any number of functions with alternating parity and increasing exponents $0 < 1 < \lambda_2 < \cdots$.

Lemma E (alternating kernels)**.** *Let $u_0, \dots, u_n$ be continuous on $\mathbb{R}$, and suppose every nontrivial combination of them has at most n distinct zeros. Given $z_1 < \cdots < z_n$, there is a nontrivial combination φ of $u_0, \dots, u_n$ with $\{\varphi = 0\} = \{z_1, \dots, z_n\}$ that has constant nonzero sign on each interval (z_j, z_{j+1}) ($z_0 = -\infty$, $z_{n+1} = +\infty$), these signs alternating in j.*

Proof. For $t_0 < \cdots < t_n$ let $D(t_0, \dots, t_n) = \det[u_i(t_j)]_{i,j=0}^n$. It never vanishes: otherwise some $c \neq 0$ satisfies $\sum_i c_i u_i(t_j) = 0$ for all j, giving a nontrivial combination with $n + 1$ zeros. Being continuous on the connected set $\{t_0 < \cdots < t_n\}$, D has a constant sign σ. Put $\varphi(x) = \det[u_i(z_1), \dots, u_i(z_n), u_i(x)]$, a combination of the u_i by expansion along the last column. It vanishes at each z_j. For $x \in (z_j, z_{j+1})$, moving the last column to position j takes $n - j$ transpositions, so $\varphi(x) = (-1)^{n-j} D(z_1, \dots, z_j, x, z_{j+1}, \dots, z_n)$, whose sign is $(-1)^{n-j}\sigma \neq 0$. In particular φ is nontrivial. □

Lemma F (sign patterns and moment constraints)**.** *Let $1 < a < r < b$, let u_i and λ_i be as in Proposition D, and let $n \in \{3, 4\}$. Let h be measurable and not a.e. zero, with $\int |h|(1 + |x|^{\lambda_n}) < \infty$ and $\int h u_i = 0$ for $i = 0, \dots, n - 1$. Suppose there are $-\infty = y_0 \leq y_1 \leq \cdots \leq y_n \leq y_{n+1} = +\infty$ such that $(-1)^{n-j} h \geq 0$ a.e. on (y_j, y_{j+1}) for $j = 0, \dots, n$ (alternating signs, the last one +). Then $\int h u_n > 0$.*

Proof. Step 1: a minimal alternating pattern. Let $J = \{j : h \text{ is not a.e. zero on } (y_j, y_{j+1})\} = \{j_1 < \cdots < j_m\}$, which is nonempty, and let $e_l = (-1)^{n-j_l}$. Group the indices of J into maximal runs of constant e_l; if j_l is the last index of a run that is not the final run, put a change point at y_{j_l+1}. These points $z_1 < \cdots < z_k$ are finite and strictly increasing: (y_{j_l}, y_{j_l+1}) has positive measure, so $y_{j_l+1} > y_{j_l} \geq -\infty$, and $y_{j_l+1} \leq y_{j_{l+1}} < y_{j_{l+1}+1} \leq +\infty$. Moreover $k \leq m - 1 \leq n$. Let ω be the step function equal to e_1 on $(-\infty, z_1)$ that switches sign at each z_i. Every interval (y_j, y_{j+1}) with $j \in J$ lies between the two change points that bound its run, so $\omega h \geq 0$ a.e., and ωh is not a.e. zero.

Step 2: $k < n$ is impossible. If $k \leq n - 1$, then $u_0, \dots, u_k$ satisfy the hypothesis of Lemma E by Proposition D. Lemma E gives $\varphi \in \operatorname{span}\{u_0, \dots, u_k\}$ with, after a choice of sign, $\varphi\omega > 0$ off $\{z_i\}$. Then $\varphi h \geq 0$ a.e. and is not a.e. zero, so $\int \varphi h > 0$; but $\int \varphi h = 0$ by the constraints.

Step 3: $k = n$. Then $m = n + 1$ and $J = \{0, \dots, n\}$, so the final sign is $e_m = +1$. Take φ from Lemma E for $u_0, \dots, u_n$ with sign chosen so that $\varphi > 0$ on (z_n, ∞). Then φ has the sign of ω off $\{z_i\}$, so $\int \varphi h > 0$. Writing $\varphi = \sum_{i \leq n} c_i u_i$, the constraints give $\int \varphi h = c_n \int h u_n$, so $c_n \neq 0$. As $x \to +\infty$, $\varphi(x)/x^{\lambda_n} \to c_n$, because $u_n(x) = x^{\lambda_n}$ for $x > 0$ has the largest exponent; since $\varphi > 0$ on (z_n, ∞), $c_n > 0$. Hence $\int h u_n > 0$. □

Lemma G (crossing pattern against Z_s)**.** *Let $X \in \mathcal{L}$ have the density $f = e^{-V} \mathbf{1}_{\operatorname{int} I}$ of Lemma A. Let $\lambda > 0$, $s \in [0, 1]$, let g be the density of λZ_s, and let $h = g - f$.*

1. *There are $-\infty \leq y_1 \leq y_2 \leq y_3 \leq y_4 \leq +\infty$ such that, a.e., h is ≥ 0, ≤ 0, ≥ 0, ≤ 0, ≥ 0 on the five successive intervals; this is the pattern of Lemma F with $n = 4$.*

2. *If $s = 1$, there are $y_1 \leq y_2 \leq y_3$ with $h \leq 0$, ≥ 0, ≤ 0, ≥ 0 a.e. on the four successive intervals; this is the pattern of Lemma F with $n = 3$.*

Proof. Put $c = 1 - s$ and $d = 1 - 2s$. For $0 < s < 1$, $Z_s = d + (sE - cE')$, and $sE - cE'$ has density $e^{-t/s}$ for $t > 0$ and $e^{t/c}$ for $t < 0$ (the tails are given in Section 3). So $g > 0$ everywhere and $\log g$ is affine on $L = (-\infty, \lambda d]$ and on $R = [\lambda d, \infty)$. For $s = 0$, $Z_0 = 1 - E'$ and g is positive and log-affine on $(-\infty, \lambda]$ and zero after; for $s = 1$, $Z_1 = E - 1$ and g is positive and log-affine on $[-\lambda, \infty)$ and zero before.

Key observation. Let P be an interval on which $g = e^{\ell}$ with ℓ affine. On $P \cap \operatorname{int} I$, $h < 0$ if and only if $\ell + V < 0$; as $\ell + V$ is convex there, this set is an interval N_P (possibly empty, possibly all of $P \cap \operatorname{int} I$, possibly half-infinite). On $P \setminus \overline{I}$, $h = g > 0$, and ∂I is null. Hence, a.e. on P, $h < 0$ exactly on the interval N_P and $h \geq 0$ on the rest of P, which consists of at most two intervals, one on each side of N_P. Where $g = 0$, $h = -f \leq 0$. Whether the endpoints of the intervals below are included is immaterial, since all statements are a.e.

The patterns follow.

- $0 < s < 1$: let (y_1, y_2) be the endpoints of N_L and (y_3, y_4) those of N_R, so $y_2 \leq \lambda d \leq y_3$. If $N_L = \emptyset$ take $y_1 = y_2 = \lambda d$; if $N_R = \emptyset$ take $y_3 = y_4 = \lambda d$.
- $s = 0$: let (y_1, y_2) be the endpoints of $N_{(-\infty,\lambda]}$, $y_3 = \lambda$ and $y_4 = +\infty$ (the fifth interval is empty); if $N_{(-\infty,\lambda]} = \emptyset$ take $y_1 = y_2 = \lambda$.
- $s = 1$, part 1: $y_1 = -\infty$, $y_2 = -\lambda$, and (y_3, y_4) the endpoints of $N_{[-\lambda,\infty)}$; if it is empty take $y_3 = y_4 = -\lambda$.
- $s = 1$, part 2: $y_1 = -\lambda$ and (y_2, y_3) the endpoints of $N_{[-\lambda,\infty)}$; if it is empty take $y_2 = y_3 = -\lambda$.

In each case the stated sign holds a.e. on each interval. □

2.2 Proofs of Theorems 3 and 4

Proof of Theorem 4. Let $Y = \lambda(E - 1) = \lambda Z_1$ with $\lambda > 0$ chosen so that $\mathbb{E}|Y|^a = \mathbb{E}|X|^a$, let g be its density and $h = g - f$. By Lemma A all moments are finite, so $\int |h|(1 + |x|^r) < \infty$. We have $\int h = 0$ (both are densities), $\int xh = 0$ (both are centred) and $\int |x|^a h = 0$ (choice of λ); these are $\int hu_i = 0$ for $i = 0, 1, 2$, with u_i as in Proposition D for any $b > r$. By Lemma G(2), h has the $n = 3$ pattern. If h is not a.e. zero, Lemma F with $n = 3$ gives $\int |x|^r \operatorname{sgn}(x)\, h > 0$, that is, $\mathbb{E}[|Y|^r \operatorname{sgn} Y] > \mathbb{E}[|X|^r \operatorname{sgn} X]$. Dividing by $\|X\|_a^r = \|Y\|_a^r$ and using the scale invariance of the ratio gives strict inequality unless $X \overset{d}{=} Y$. □

Proof of Theorem 3. Fix $r = (a+b)/2$ and let $\rho(Y) = \mathbb{E}[|Y|^r \operatorname{sgn} Y]/(\mathbb{E}|Y|^a)^{r/a}$. It is scale invariant and $\rho(-Y) = -\rho(Y)$. Theorem 4 applied to X and to $-X$ (both in $\mathcal{L}$) gives $\rho(X) \leq \rho(Z_1)$ and $-\rho(X) \leq \rho(Z_1)$. Since $Z_0 \overset{d}{=} -Z_1$, $\rho(Z_0) = -\rho(Z_1)$, so

$$\rho(Z_0) \leq \rho(X) \leq \rho(Z_1).$$

The map $s \mapsto Z_s$ is pointwise continuous and $|Z_s| \leq E + E' + 2$. For each $t \geq 0$, $(E + E' + 2)^t$ is integrable, so by dominated convergence $s \mapsto \mathbb{E}|Z_s|^t$ and $s \mapsto \mathbb{E}[|Z_s|^r \operatorname{sgn} Z_s]$ are continuous on $[0, 1]$; moreover $\mathbb{E}|Z_s|^t > 0$. Hence $s \mapsto \rho(Z_s)$ is continuous, and the intermediate value theorem gives $s_0 \in [0, 1]$ with $\rho(Z_{s_0}) = \rho(X)$.

Let $\lambda = \|X\|_a / \|Z_{s_0}\|_a$ and $Y = \lambda Z_{s_0}$, with density g_Y. Then $\mathbb{E}|Y|^a = \mathbb{E}|X|^a$ and $\mathbb{E}[|Y|^r \operatorname{sgn} Y] = \rho(X)(\mathbb{E}|X|^a)^{r/a} = \mathbb{E}[|X|^r \operatorname{sgn} X]$. With $h = g_Y - f$ we have $\int |h|(1 + |x|^b) < \infty$ and $\int hu_i = 0$ for $i = 0, \ldots, 3$. By Lemma G(1), h has the $n = 4$ pattern, and Lemma F gives $\mathbb{E}|Y|^b - \mathbb{E}|X|^b = \int |x|^b h > 0$ unless $h = 0$ a.e., that is, unless $X \overset{d}{=} Y$.

Consequences. Let $Q(s) = \|Z_s\|_b/\|Z_s\|_a$. By the continuity just proved (with $t = a$ and $t = b$), Q is continuous on the compact interval $[0,1]$ and attains its maximum. Then

$$\frac{\|X\|_b}{\|X\|_a} \le \frac{\|Y\|_b}{\|Y\|_a} = Q(s_0) \le \max Q.$$

If $\|X\|_b/\|X\|_a = \max Q$, both inequalities are equalities: the first forces $X \overset{d}{=} \lambda Z_{s_0}$, and the second shows that s_0 is a maximiser. Conversely, if $X \overset{d}{=} \lambda Z_{s_1}$ with s_1 a maximiser, equality is immediate. □

Remark 5. If $s_0 \in \{0,1\}$ in the proof of Theorem 3, then h has at most three sign changes, and Lemma F forces $h = 0$ a.e. For $s_0 = 1$, the pattern of Lemma G(2) $(-,+,-,+)$ is the $n = 4$ pattern of Lemma F with an empty first interval, and Step 2 of the proof of Lemma F ($k \le 3 < 4$) rules out $h \ne 0$. For $s_0 = 0$ the pattern is $+,-,+,-$, so Lemma F is applied in the same way to $-h$, which satisfies the same four constraints. This is consistent with the equality case of Theorem 4.

3 Reduction to a two-parameter inequality

Each Z_s is itself in $\mathcal{L}$ (for $0 < s < 1$ it has the density of an asymmetric Laplace law shifted to mean zero), so the maximum in Theorem 3 is the sharp constant. Applying Theorem 3 with $(a,b) = (p,2)$ for $1 < p < 2$ and $(2,p)$ for $p > 2$, and using $Z_{1-s} \overset{d}{=} -Z_s$:

$$\log m_p = \min_{0\le s\le 1/2} G(s,p) \quad (1<p<2), \qquad \log M_p = \max_{0\le s\le 1/2} G(s,p) \quad (p>2),$$

where $G(s,p) = \frac{1}{p}\log M(s,p) - \frac{1}{2}\log N(s)$, $M(s,p) = \mathbb{E}|Z_s|^p$ and $N(s) = \mathbb{E}Z_s^2 = s^2 + (1-s)^2$. Since $G(0,p) = \frac{1}{p}\log F(p) =: G_0(p)$ and $G(\frac{1}{2},p) = \frac{1}{p}\log\Gamma(p+1) - \frac{1}{2}\log 2 =: G_h(p)$, Theorem 1 is equivalent to:

(T_A) $G(s,p) \ge G_h(p)$ for $1 < p \le p^*$, $0 \le s \le \frac{1}{2}$;

(T_B) $G(s,p) \ge G_0(p)$ for $p^* \le p < 2$, $0 \le s \le \frac{1}{2}$;

(T_C) $G(s,p) \le G_0(p)$ for $p > 2$, $0 \le s \le \frac{1}{2}$;

together with the existence and uniqueness of p^*.

A representation. Put $c = 1-s$, $d = 1-2s$, and let R be standard exponential. For $0 < s \le \frac{1}{2}$, $sE - cE'$ has tails $\mathbb{P}(sE - cE' > t) = se^{-t/s}$ and $\mathbb{P}(sE - cE' < -t) = ce^{-t/c}$ ($t \ge 0$), the same as the mixture equal to sR with probability s and $-cR$ with probability c. Since $Z_s = sE - cE' + d$,

$$M(s,p) = s\,\mathbb{E}(d + sR)^p + c\,\mathbb{E}|d - cR|^p, \tag{1}$$

and $\mathbb{E}|x-R|^p = e^{-x}\big(\Gamma(p+1) + \int_0^x u^p e^u\,du\big)$ for $x \ge 0$. In particular $F(p) = e^{-1}\big(\Gamma(p+1) + \int_0^1 u^p e^u\,du\big)$.

4 Large exponents

Proposition 6. *For every real $p \geq 7$ and $0 < s \leq \frac{1}{2}$,*

$$N(s)^{p/2}F(p) - M(s,p) \;\geq\; \frac{s^2(1-s)^{p-1}}{e}\,K(p) > 0,$$
$$K(p) = \Big(\frac{p}{2} - 2e + 2\Big)\Gamma(p+1) - \frac{e(p-2)}{2} - \frac{e(e-2)}{p+1}.$$

In particular $G(s,p) < G_0(p)$, so (T_C) holds for $p \geq 7$.

Proof. Write $\Gamma_p = \Gamma(p+1)$ and $J(x) = \int_0^x u^p e^u\,du$. Put $\tau = s/c \in (0,1]$, so that $c = 1/(1+\tau)$, $d/c = 1-\tau$ and $N(s) = c^2(1+\tau^2)$. By (1),

$$M(s,p) = \frac{c^p}{1+\tau}\Big[\tau\,\mathbb{E}(1-\tau+\tau R)^p + e^{\tau-1}\big(\Gamma_p + J(1-\tau)\big)\Big],$$
$$N(s)^{p/2}F(p) = c^p(1+\tau^2)^{p/2}e^{-1}\big(\Gamma_p + J(1)\big).$$

With $A(\tau) = (1+\tau)(1+\tau^2)^{p/2}$ this gives $e(1+\tau)c^{-p}\big[N^{p/2}F - M\big] = S(\tau)$, where

$$S(\tau) = A(\tau)\big(\Gamma_p + J(1)\big) - e\tau\,\mathbb{E}(1-\tau+\tau R)^p - e^{\tau}\big(\Gamma_p + J(1-\tau)\big).$$

Since $t \mapsto t^p$ is convex and $1-\tau+\tau R$ is a convex combination of 1 and R, $\mathbb{E}(1-\tau+\tau R)^p \leq (1-\tau) + \tau\Gamma_p$. Hence $S \geq \Delta$ with

$$\Delta(\tau) = \big[A(\tau) - e^{\tau} - e\tau^2\big]\Gamma_p + \Big\{A(\tau)J(1) - e^{\tau}J(1-\tau) - e\tau(1-\tau)\Big\}.$$

For $\tau \in [0,1]$ and $p \geq 2$ we use:

(a) $A(\tau) \geq (1+\tau)(1+\frac{p}{2}\tau^2) \geq 1 + \tau + \frac{p}{2}\tau^2$, by Bernoulli's inequality with exponent $p/2 \geq 1$;

(b) $0 \leq e^{\tau} - 1 - \tau \leq (e-2)\tau^2$;

(c) $0 \leq J(1) \leq e/(p+1)$;

(d) $\int_{1-\tau}^1 u^p e^u\,du \geq e^{1-\tau}\big(1-(1-\tau)^{p+1}\big)/(p+1)$;

(e) $1-(1-\tau)^{p+1} \geq (p+1)\tau - \frac{(p+1)p}{2}\tau^2$, by Taylor's theorem, since the third derivative of $(1-\tau)^{p+1}$ is ≤ 0 on $[0,1]$.

By (a) and (b), the first bracket is at least $(\frac{p}{2} - 2e + 2)\tau^2$. For the second, write $J(1-\tau) = J(1) - \int_{1-\tau}^1 u^p e^u\,du$ and use (a), (b), (c), (d), (e):

$$A(\tau)J(1) - e^{\tau}J(1-\tau) \geq (1+\tau-e^{\tau})J(1) + e\,\frac{1-(1-\tau)^{p+1}}{p+1} \geq -\frac{e(e-2)}{p+1}\tau^2 + e\Big(\tau - \frac{p}{2}\tau^2\Big).$$

Subtracting $e\tau(1-\tau)$, the terms linear in τ cancel, and the second bracket is at least $-\big(\frac{e(e-2)}{p+1} + \frac{e(p-2)}{2}\big)\tau^2$. Hence $S(\tau) \geq \Delta(\tau) \geq \tau^2 K(p)$, which is the stated bound after multiplying by $c^p/(e(1+\tau))$.

Finally let $p \geq 7$. Then $\frac{p}{2} - 2e + 2 > \frac{11}{2} - \frac{87}{16} = \frac{1}{16}$ (as $e < \frac{87}{32}$). Since Γ is increasing on $[2,\infty)$, $\Gamma_p = p\,\Gamma(p) \geq 720p$. Also $\frac{e(p-2)}{2} < \frac{87}{64}p$ and $\frac{e(e-2)}{p+1} < \frac{87}{32}\cdot\frac{23}{32}\cdot\frac{1}{8} = \frac{2001}{8192}$. Thus

$$K(p) > 45p - \frac{87}{64}p - \frac{2001}{8192} = \frac{2793}{64}p - \frac{2001}{8192} > 0. \qquad \square$$

Remark 7. The argument needs $\frac{p}{2} > 2e - 2$, i.e. $p > 4e - 4 \approx 6.87$; indeed $K(6.8) < 0 < K(6.9)$. The loss comes from the chord bound for $\mathbb{E}(1-\tau+\tau R)^p$.

5 Derivatives in s and the endpoints

Let $W = E + E' - 2 = \partial_s Z_s$.

Lemma 8. *Let $0 < s < 1$ and $X = sE - cE'$, so that $Z_s = d + X$. Then $\mathbb{P}(X > 0) = s$, and conditionally on $\{X > 0\}$ the pair (Z_s, W) has the law of $(d + sR,\ (R-1)+(T-1))$. Conditionally on $\{X < 0\}$ it has the law of $(d - cR,\ (R-1)+(T-1))$. Here R, T are independent standard exponential random variables. At $s = 0, 1$ use the direct formulas for Z_s, W, or their moment limits; no conditioning on a zero-probability event is intended.*

Proof. On $\{sE > cE'\}$, condition on E'. By memorylessness $R := E - cE'/s$ is standard exponential and independent of E', while E' has conditional density proportional to $e^{-u}e^{-cu/s} = e^{-u/s}$ at u, so $E' = sT$. Then $X = sR$ and $W = R + cT + sT - 2$. The other side is symmetric. □

Lemma 9. *For $p > 1$, $G(\cdot, p)$ is C^2 on $[0, \frac{1}{2}]$. With $\tau = -d/N$ and $V = W - \tau Z_s$, put $A = \mathbb{E}[|Z_s|^{p-2} Z_s V]/M$ and $B = \mathbb{E}[|Z_s|^{p-2} V^2]/M$. Then*

$$\partial_s G = A, \qquad \partial_s^2 G = (p-1)B - pA^2 + \frac{2d}{N}A - \frac{1}{N^2}.$$

Proof. The map $s \mapsto Z_s$ is affine with slope W. Differentiation under the expectation is justified by dominated convergence for the first derivative. For the second, use the absolute continuity of $z \mapsto |z|^{p-2}z$, whose derivative $(p-1)|z|^{p-2}$ is locally integrable for $p > 1$, together with Tonelli; finiteness comes from the closed forms in supplement `local.md`. This gives $M_s = p\mathbb{E}[|Z|^{p-2} ZW]$ and $M_{ss} = p(p-1)\mathbb{E}[|Z|^{p-2} W^2]$. Substituting $W = V + \tau Z$ and using $\tau' + \tau^2 = 1/N^2$ (as $d^2 + 4sc = 1$) gives the formulas. □

Proposition 10. *For every $p > 1$:*

(i) $\partial_s G(0, p) = 0$ and $\partial_s G(\frac{1}{2}, p) = 0$;

(ii) $\partial_s^2 G(0, p) = \frac{p-1}{p}\big(\frac{1}{F(p)} - 1\big)$ and $\partial_s^2 G(\frac{1}{2}, p) = \frac{4(p-2)^2}{p}$.

Proof. At $s = 0$, $c = d = 1$ and $V = W + Z = E - 1$ is independent of $Z = 1 - E'$, so $\mathbb{E}[|Z|^{p-2} ZV] = 0$. Hence $A = 0$ and $B = F(p-2)/F(p)$. Two integrations by parts against e^{-t} give $p(p-1)F(p-2) = F(p) - 1 + p$, which yields the first formula in (ii).

At $s = \frac{1}{2}$, $Z_{1-s} \overset{d}{=} -Z_s$ (exchange E and E') and $N(1-s) = N(s)$, so $G(\cdot, p)$ is even about $\frac{1}{2}$, and $\partial_s G(\frac{1}{2}, p) = 0$. There $d = 0$, $N = \frac{1}{2}$, $M = 2^{-p}\Gamma(p+1)$, $|Z| = R/2$ on both sides, and $\mathbb{E}[V \mid \cdot] = R - 1$. So $(p-1)B = 4(p-1)\big(\Gamma(p+1) - 2\Gamma(p) + 2\Gamma(p-1)\big)/\Gamma(p+1) = 4(p - 3 + 4/p)$, and $\partial_s^2 G = 4(p - 3 + 4/p) - 4 = 4(p-2)^2/p$. □

Proposition 11 (computer-assisted)**.** *(L1) $\partial_s^2 G > \frac{6}{125}$ on $[\frac{9}{20}, \frac{1}{2}] \times (1, \frac{7}{4}]$.*

(L2) $\partial_s^2 G > \frac{29}{2000}$ on $[0, \frac{1}{20}] \times [\frac{8}{5}, \frac{39}{20}]$.

(L3) $\partial_s^2 G < -\frac{31}{2000}$ on $[0, \frac{1}{20}] \times [\frac{41}{20}, 7]$.

Consequently, $G \geq G_h$ on $[\frac{9}{20}, \frac{1}{2}] \times (1, \frac{7}{4}]$, $G \geq G_0$ on $[0, \frac{1}{20}] \times [\frac{8}{5}, \frac{39}{20}]$, and $G \leq G_0$ on $[0, \frac{1}{20}] \times [\frac{41}{20}, 7]$.

Certificate. By Lemma 8, the quantities M, $\mathbb{E}[|Z|^{p-2} ZV]$ and $(p-1)\mathbb{E}[|Z|^{p-2} V^2]$ are explicit combinations of $\Gamma(p+1)$, $\Gamma(p)$, upper incomplete gamma functions, $\int_0^x u^a e^u\, du = \frac{x^{a+1}}{a+1}\,{}_1F_1(a+1; a+2; x)$ and ${}_1F_1(1; b; z)$ (supplement `local.md`, Lemma 4 and Proposition B). The terms that are singular

as $p \to 1$ appear with a factor $p-1$ and are rewritten as $d^{p-1}\big[\frac{N}{c^2}{}_1F_1(1;p;-x) - \frac{N}{s^2}{}_1F_1(1;p;d/s)\big]$, which is continuous at the corner $(\frac{1}{2}, 1)$, where $\partial_s^2 G = 4$.

Each region is covered by boxes with exact rational endpoints. On each box, p is a ball, the s-dependence is enclosed through monotone hulls of the special-function pieces, and d^{p-1} is enclosed in $[d_0^{e_1}, d_1^{e_0}]$. Precision is 128 bits, boxes are bisected down to width 2^{-22}, and an undecided comparison counts as a failure. Region L1 used 50,214 boxes, L2 129,875 and L3 57,496, all without failure.

The consequences follow from Proposition 10(i) and Taylor's theorem: $\partial_s G$ has a fixed sign on each strip away from the endpoint. □

6 The neighbourhood of $p = 2$

At $p = 2$ we have $M(s,2) = N(s)$, so $G(\cdot, 2) \equiv 0$ and every comparison degenerates. Let $\mathcal{H}(s,p) = \partial_p\big[G(s,p) - G_0(p)\big]$.

Proposition 12 (computer-assisted)**.** *(S1)* $\mathcal{H} < 0$ *on* $[\frac{1}{20}, \frac{1}{2}] \times [\frac{39}{20}, \frac{41}{20}]$.

(S2) $\partial_p \partial_s^2 G < 0$ *on* $[0, \frac{1}{20}] \times [\frac{39}{20}, \frac{41}{20}]$.

Consequently, for $p \in [\frac{39}{20}, \frac{41}{20}]$ *and* $0 < s \le \frac{1}{2}$, $G(s,p) - G_0(p)$ *has the sign of* $2 - p$.

Proof. Consequence. Since $G(0,\cdot) = G_0$, $\mathcal{H}(0,p) = 0$. The mixed partial derivatives of G up to $\partial_p \partial_s^2$ exist, are continuous and commute on the strip (supplement `strip.md`, Lemma 3), so $\partial_s \mathcal{H}(0,p) = \partial_p \partial_s G(0,p) = 0$ by Proposition 10(i), and $\partial_s^2 \mathcal{H} = \partial_p \partial_s^2 G < 0$ on $[0, \frac{1}{20}]$ by (S2), so Taylor's formula gives $\mathcal{H} < 0$ on $(0, \frac{1}{20}]$. Together with (S1), $\mathcal{H} < 0$ for $0 < s \le \frac{1}{2}$. Since $G(s,2) - G_0(2) = 0$, we have $G(s,p) - G_0(p) = \int_2^p \mathcal{H}(s,t)\,dt$, which has the sign of $2 - p$.

Certificate. Write $q = p - 2$, $\Phi_A = \mathbb{E}|Z_s|^{2+q}$, $\Phi_B = \mathbb{E}[|Z_s|^q Z_s(R-1)]$ and $\Phi_C = \mathbb{E}[|Z_s|^q((R-1)^2+1)]$, the expectations being taken in the conditioned form of Lemma 8. Then $M_s = p\Phi_B$, $M_{ss} = p(p-1)\Phi_C$, $\partial_s \partial_p G = \partial_q(\Phi_B/\Phi_A)$ and $\partial_p \partial_s^2 G = \partial_q\big[(1+q)\Phi_C/\Phi_A - (2+q)(\Phi_B/\Phi_A)^2\big]$ (with the N-terms independent of p).

On the negative side ($Z \le d$) the moments have closed forms in q via Gamma functions and a rapidly convergent series with bounded tail. On the positive side they are enclosed by order-12 Taylor polynomials in q, whose coefficients are rigorous integrals (Arb's `acb.integral` with explicit tail and endpoint bounds), plus an explicit remainder bound.

S1 is certified with a mean-value form in s and q (157 boxes, worst certified upper bound below -1.5×10^{-5}). S2 is certified with direct evaluation in s and a mean-value form in q (64 boxes, worst bound below -0.040). The shipped outputs retain exact rational enclosure endpoints. Both certificates are regenerated in normal and optimised replay; the regression suite rejects altered signs, empty or overlapping coverage, and malformed domains. □

7 The crossing point and the bulk

Let $H(p) = G_0(p) - G_h(p)$, where $G_0(p) = \frac{1}{p}\log F(p)$ and $G_h(p) = \frac{1}{p}\log\Gamma(p+1) - \frac{1}{2}\log 2$.

Proposition 13 (computer-assisted)**.** *Let* $p^*_{\mathrm{lo}} = 1.682457208$ *and* $p^*_{\mathrm{hi}} = 1.682457209$. *Then:*

- $H > 0$ *on* $[1, p^*_{\mathrm{lo}}]$;
- $H < 0$ *on* $[p^*_{\mathrm{hi}}, \frac{39}{20}]$;
- $H' < -0.016$ *on* $[p^*_{\mathrm{lo}}, p^*_{\mathrm{hi}}]$.

Hence H has exactly one zero p^* in $[1, \frac{39}{20}]$, $H > 0$ on $[1, p^*)$, and $H < 0$ on $(p^*, \frac{39}{20}]$.

Proposition 14 (computer-assisted)**.** *Let $\eta = \frac{1}{20}$. The following hold:*

(B1) $G(s,p) > G_h(p)$ for $s \in [0, \frac{9}{20}]$, $p \in [1, p^*_{\mathrm{hi}} - \eta]$;

(B2) $G(s,p) > \min\{G_0(p), G_h(p)\}$ for $s \in [\frac{1}{20}, \frac{9}{20}]$, $p \in [p^*_{\mathrm{lo}} - \eta, p^*_{\mathrm{hi}} + \eta]$;

(B3) $G(s,p) > G_0(p)$ for $s \in [\frac{1}{20}, \frac{1}{2}]$, $p \in [p^*_{\mathrm{lo}} + \eta, \frac{39}{20}]$;

(B4) $G(s,p) < G_0(p)$ for $s \in [\frac{1}{20}, \frac{1}{2}]$, $p \in [\frac{41}{20}, 7]$.

Certificates. $M(s,p) = s\,J_p + c^{p+1}e^{-x}\big(\Gamma(p+1) + \frac{x^{p+1}}{p+1}{}_1F_1(p+1; p+2; x)\big)$, where $x = d/c$ and $J_q = s^q e^{d/s}\Gamma(q+1, d/s)$. The partial derivatives M_s and $M_p = \mathbb{E}|Z_s|^p \log|Z_s|$ have explicit forms (supplement `bulk.md`).

On a box with centre (s_0, p_0) and half-widths (h_s, h_p), the difference D to be certified is enclosed by $D(s_0,p_0) + [\partial_p D]\,[-h_p, h_p] + [\partial_s D]\,[-h_s, h_s]$. Here $[\partial_p D]$ is bounded through the convexity of $p \mapsto \mathbb{E}|Z_s|^p$: $M_p = \mathbb{E}|Z_s|^p \log|Z_s|$ and F' are nondecreasing in p (their derivatives are $\mathbb{E}|Z_s|^p \log^2|Z_s| \geq 0$), so their values at the exact rational endpoints bound them; M and F are enclosed separately and interval division is used. $[\partial_s D]$ is bounded from monotone enclosures of the closed-form pieces in s. No incomplete gamma function is evaluated at a wide parameter ball.

Branch and bound at 128 bits gives:

- B1: 1653 boxes, margin $\geq 3.2 \times 10^{-6}$;
- B2: 1127 boxes, margin $\geq 4.0 \times 10^{-8}$;
- B3: 4413 boxes, margin $\geq 1.4 \times 10^{-8}$;
- B4: 8635 boxes, margin $\geq 8.0 \times 10^{-8}$;
- the crossing lemma: 73, 65 and 1 boxes.

There were no failures. B1–B4 are regenerated at 128 bits in both normal and optimised replay. The four deliberately false bulk statements are rejected in both modes. Optional floating-point quadrature checks are diagnostic only and do not replace the interval certificates. □

8 Proof of Theorem 1

By Section 3 it suffices to prove (T_A), (T_B), (T_C) and the uniqueness of p^*. Throughout, $s \in [0, \frac{1}{2}]$.

The crossing. By Proposition 13, $H = G_0 - G_h$ has a unique zero p^* in $[1, \frac{39}{20}]$, with $H > 0$ before it and $H < 0$ after it. On $[\frac{39}{20}, 2)$, Proposition 12 at $s = \frac{1}{2}$ gives $G_h - G_0 > 0$, so p^* is the unique zero in $(1, 2)$.

($\mathbf{T}_A$): $1 < p \leq p^*$.

- For $p \leq p^*_{\mathrm{hi}} - \eta$: B1 covers $s \leq \frac{9}{20}$ and L1 covers $s \geq \frac{9}{20}$.
- For $p \in [p^*_{\mathrm{hi}} - \eta, p^*]$:
  - for $s \leq \frac{1}{20}$, L2 gives $G \geq G_0$, and $G_0 \geq G_h$ because $p \leq p^*$;
  - for $s \in [\frac{1}{20}, \frac{9}{20}]$, B2 gives $G > \min\{G_0, G_h\} = G_h$;
  - for $s \geq \frac{9}{20}$, L1 applies.

($\mathbf{T}_B$): $p^* \le p < 2$.

- For $p \in [p^*, p^*_{\mathrm{lo}} + \eta]$:
  - L2 covers $s \le \frac{1}{20}$;
  - B2 gives $G > \min\{G_0, G_h\} = G_0$ on $[\frac{1}{20}, \frac{9}{20}]$;
  - L1 gives $G \ge G_h \ge G_0$ on $[\frac{9}{20}, \frac{1}{2}]$.
- For $p \in [p^*_{\mathrm{lo}} + \eta, \frac{39}{20}]$: L2 covers $s \le \frac{1}{20}$ and B3 the rest.
- For $p \in [\frac{39}{20}, 2)$: Proposition 12.

($\mathbf{T}_C$): $p > 2$.

- $(2, \frac{41}{20}]$: Proposition 12.
- $[\frac{41}{20}, 7]$: L3 for $s \le \frac{1}{20}$ and B4 for the rest.
- $[7, \infty)$: Proposition 6. □

9 Equality cases: proof of Theorem 2

Reduction to G. Recall $G(1-s,p) = G(s,p)$, since $Z_{1-s} \overset{d}{=} -Z_s$ and $N(1-s) = N(s)$. For $1 < p < 2$ apply Theorem 3 with $(a,b) = (p,2)$, where $\|Z_s\|_2/\|Z_s\|_p = e^{-G(s,p)}$; for $p > 2$ apply it with $(a,b) = (2,p)$, where $\|Z_s\|_p/\|Z_s\|_2 = e^{G(s,p)}$. By the equality clause of Theorem 3, equality in Theorem 1 holds for $X \in \mathcal{L}$ if and only if $X \overset{d}{=} \lambda Z_{s_1}$ for some $\lambda > 0$ and some $s_1 \in [0,1]$ that minimises $G(\cdot,p)$ on $[0,1]$ when $1 < p < 2$, or maximises it when $p > 2$. Theorem 2 therefore follows from Lemma S below together with this symmetry. Lemma S shows that the set of extremal parameters in $[0,1]$ is $\{\frac{1}{2}\}$ in case (i), $\{0, \frac{1}{2}, 1\}$ in case (ii), and $\{0,1\}$ in cases (iii) and (iv). Finally, $\lambda Z_{1/2} = \frac{\lambda}{2}(E - E')$, $\lambda Z_1 = \lambda(E-1)$ and $\lambda Z_0 \overset{d}{=} -\lambda(E-1)$.

Facts used. Lemma S uses only the following statements, all established above with strict inequalities on the stated closed sets. Here $\eta = \frac{1}{20}$, $p^*_{\mathrm{lo}} = 1.682457208$, $p^*_{\mathrm{hi}} = 1.682457209$, and G_0, G_h, $H = G_0 - G_h$ are as in Sections 3 and 7.

(F1) Lemma 9: for each $p > 1$, $G(\cdot,p)$ is C^2 on $[0, \frac{1}{2}]$.

(F2) Proposition 10(i): for each $p > 1$, $\partial_s G(0,p) = \partial_s G(\frac{1}{2}, p) = 0$.

(F3) Proposition 11: (L1) $\partial_s^2 G > \frac{6}{125}$ on $[\frac{9}{20}, \frac{1}{2}] \times (1, \frac{7}{4}]$; (L2) $\partial_s^2 G > \frac{29}{2000}$ on $[0, \frac{1}{20}] \times [\frac{8}{5}, \frac{39}{20}]$; (L3) $\partial_s^2 G < -\frac{31}{2000}$ on $[0, \frac{1}{20}] \times [\frac{41}{20}, 7]$.

(F4) Proposition 12: for $p \in [\frac{39}{20}, \frac{41}{20}]$ and $0 < s \le \frac{1}{2}$, $G(s,p) - G_0(p) = \int_2^p \mathcal{H}(s,t)\,dt$ with $\mathcal{H} < 0$; hence $G(s,p) > G_0(p)$ for $\frac{39}{20} \le p < 2$ and $G(s,p) < G_0(p)$ for $2 < p \le \frac{41}{20}$.

(F5) Proposition 13: $H > 0$ on $[1, p^*_{\mathrm{lo}}]$, $H < 0$ on $[p^*_{\mathrm{hi}}, \frac{39}{20}]$ and $H' < 0$ on $[p^*_{\mathrm{lo}}, p^*_{\mathrm{hi}}]$. Hence $p^*_{\mathrm{lo}} < p^* < p^*_{\mathrm{hi}}$, $H > 0$ on $[1, p^*)$, $H(p^*) = 0$ and $H < 0$ on $(p^*, \frac{39}{20}]$.

(F6) Proposition 14, on closed rectangles: (B1) $G > G_h$ on $[0, \frac{9}{20}] \times [1, p^*_{\mathrm{hi}} - \eta]$; (B2) $G > \min\{G_0, G_h\}$ on $[\frac{1}{20}, \frac{9}{20}] \times [p^*_{\mathrm{lo}} - \eta, p^*_{\mathrm{hi}} + \eta]$; (B3) $G > G_0$ on $[\frac{1}{20}, \frac{1}{2}] \times [p^*_{\mathrm{lo}} + \eta, \frac{39}{20}]$; (B4) $G < G_0$ on $[\frac{1}{20}, \frac{1}{2}] \times [\frac{41}{20}, 7]$. The certified minimum margins are 3.2×10^{-6}, 4.0×10^{-8}, 1.4×10^{-8} and 8.0×10^{-8}.

(F7) Proposition 6: for $p \geq 7$ and $0 < s \leq \frac{1}{2}$, $N(s)^{p/2}F(p) - M(s,p) > 0$. Since $G(s,p) - G_0(p) = \frac{1}{p}\log\big(M(s,p)/(N(s)^{p/2}F(p))\big)$, this is $G(s,p) < G_0(p)$.

Lemma S (strict separation)**.** *For $s \in [0, \frac{1}{2}]$:*

(a) if $1 < p < p^*$, then $G(s,p) > G_h(p)$ for $s < \frac{1}{2}$;

(b) if $p = p^*$, then $G(s,p^*) > G_0(p^*) = G_h(p^*)$ for $0 < s < \frac{1}{2}$;

(c) if $p^* < p < 2$, then $G(s,p) > G_0(p)$ for $s > 0$;

(d) if $p > 2$, then $G(s,p) < G_0(p)$ for $s > 0$.

Proof. Strict local consequences. By (F1), (F2) and Taylor's formula with integral remainder, (F3) gives:

- on L1, for $\frac{9}{20} \leq s < \frac{1}{2}$ and $1 < p \leq \frac{7}{4}$: $G(s,p) - G_h(p) = \int_s^{1/2}(t-s)\,\partial_s^2 G(t,p)\,dt \geq \frac{3}{125}(\frac{1}{2}-s)^2 > 0$;
- on L2, for $0 < s \leq \frac{1}{20}$ and $\frac{8}{5} \leq p \leq \frac{39}{20}$: $G(s,p) - G_0(p) = \int_0^s (s-t)\,\partial_s^2 G(t,p)\,dt \geq \frac{29}{4000}s^2 > 0$;
- on L3, for $0 < s \leq \frac{1}{20}$ and $\frac{41}{20} \leq p \leq 7$: $G(s,p) - G_0(p) \leq -\frac{31}{4000}s^2 < 0$.

Note $\frac{8}{5} < p^*_{\mathrm{hi}} - \eta < p^*_{\mathrm{lo}} + \eta < \frac{7}{4} < \frac{39}{20}$.

(a) $1 < p < p^*$. If $p \leq p^*_{\mathrm{hi}} - \eta$, B1 gives $G > G_h$ for $s \leq \frac{9}{20}$ and L1 for $\frac{9}{20} \leq s < \frac{1}{2}$. If $p \in [p^*_{\mathrm{hi}} - \eta, p^*)$: at $s = 0$, $G = G_0 > G_h$ by (F5); for $0 < s \leq \frac{1}{20}$, L2 gives $G > G_0 > G_h$; for $s \in [\frac{1}{20}, \frac{9}{20}]$, B2 gives $G > \min\{G_0, G_h\} = G_h$; for $s \in [\frac{9}{20}, \frac{1}{2})$, L1 applies.

(b) $p = p^*$. $G_0(p^*) = G_h(p^*)$ by (F5). For $0 < s \leq \frac{1}{20}$, L2 gives $G > G_0(p^*)$; for $s \in [\frac{1}{20}, \frac{9}{20}]$, B2 gives $G > \min\{G_0, G_h\}(p^*)$; for $s \in [\frac{9}{20}, \frac{1}{2})$, L1 gives $G > G_h(p^*)$.

(c) $p^* < p < 2$. If $p \in (p^*, p^*_{\mathrm{lo}} + \eta]$: for $0 < s \leq \frac{1}{20}$, L2; for $s \in [\frac{1}{20}, \frac{9}{20}]$, B2 gives $G > \min\{G_0, G_h\} = G_0$ since $H < 0$; for $s \in [\frac{9}{20}, \frac{1}{2}]$, L1 gives $G \geq G_h > G_0$. If $p \in [p^*_{\mathrm{lo}} + \eta, \frac{39}{20}]$: L2 for $0 < s \leq \frac{1}{20}$ and B3 for $s \geq \frac{1}{20}$. If $p \in [\frac{39}{20}, 2)$: (F4).

(d) $p > 2$. If $p \in (2, \frac{41}{20}]$: (F4). If $p \in [\frac{41}{20}, 7]$: L3 for $0 < s \leq \frac{1}{20}$ and B4 for $s \in [\frac{1}{20}, \frac{1}{2}]$. If $p \geq 7$: (F7).

The region boundaries are those of the assembly in Section 8. □

Remark 15. The same bounds give quadratic lower bounds in s near the extremal parameters; for example $G(s,p) - G_0(p) \geq \frac{29}{4000}s^2$ on $[0, \frac{1}{20}] \times [\frac{8}{5}, \frac{39}{20}]$. Combined with Theorem 3, this is a stability statement for the reduced parameter s_0 only. It is not stability in law, and it is not claimed as such.

10 Qualitative stability in distribution

The equality classification has the following compactness consequence. The convergence results used here are standard; no quantitative deficit-to-distance rate is asserted.

Let $\mathcal{C}$ be the laws of mean-zero, variance-one log-concave random variables. Write

$$\nu_L = \mathrm{Law}((E - E')/\sqrt{2}), \qquad \nu_+ = \mathrm{Law}(E - 1), \qquad \nu_- = \mathrm{Law}(1 - E),$$

and define the variance-normalised equality set

$$\mathcal{A}_p = \begin{cases} \{\nu_L\}, & 1 < p < p^*, \\ \{\nu_L, \nu_+, \nu_-\}, & p = p^*, \\ \{\nu_+, \nu_-\}, & p^* < p < 2 \text{ or } p > 2. \end{cases}$$

For X with law $\mu \in \mathcal{C}$, put

$$\Delta_p(\mu) = \begin{cases} \log(\|X\|_p / m_p), & 1 < p < 2, \\ \log(M_p / \|X\|_p), & p > 2. \end{cases}$$

This is nonnegative by Theorem 1. For $1 \le q < \infty$, use the conventions

$$d_{\mathrm{TV}}(\mu,\nu) = \sup_{B \text{ Borel}} |\mu(B) - \nu(B)|, \qquad W_q(\mu,\nu) = \inf_{(X,Y)} (\mathbb{E}|X-Y|^q)^{1/q},$$

where the infimum is over couplings with laws μ, ν, and set $d_q(\mu, \mathcal{A}) = \min_{\nu \in \mathcal{A}} [d_{\mathrm{TV}}(\mu,\nu) + W_q(\mu,\nu)]$ for finite nonempty $\mathcal{A}$.

Corollary 16. *Fix $p > 1$, $p \neq 2$, and $1 \le q < \infty$. There is a finite, nondecreasing function $\omega_{p,q} : [0,\infty) \to [0,\infty)$ with $\omega_{p,q}(t) \to 0$ as $t \downarrow 0$ such that*

$$d_q(\mu, \mathcal{A}_p) \le \omega_{p,q}(\Delta_p(\mu)) \qquad (\mu \in \mathcal{C}).$$

Moreover, if K is a nonempty compact subset of one of $(1, p^*)$, $(p^*, 2)$ or $(2, \infty)$, a single function $\omega_{K,q}$ has this property simultaneously for every $p \in K$.

Proof. We first record a uniform tail bound. For X with law in $\mathcal{C}$, let $S(t) = \Pr(X \ge t)$. This function is log-concave: apply preservation under marginalisation to $f(t+u)\mathbf{1}_{\{u \ge 0\}}$, where f is the density of X; see [10, Theorem 3.3]. Chebyshev's inequality gives $S(2) \le 1/4$ and $S(-2) \ge 3/4$. If $S(2) > 0$, the decreasing secant slopes of $\log S$ imply, for $t \ge 2$,

$$S(t) \le S(2) \left(\frac{S(2)}{S(-2)} \right)^{(t-2)/4} \le \frac{1}{4}\, 3^{-(t-2)/4}.$$

The bound is immediate if $S(2) = 0$, and also holds where $S(t) = 0$. Applying it to $-X$ gives

$$\sup_{\mu \in \mathcal{C}} \Pr_{\mu}(|X| \ge t) \le \frac{1}{2}\, 3^{-(t-2)/4} \qquad (t \ge 2). \tag{2}$$

In particular every fixed absolute moment is uniformly integrable on $\mathcal{C}$.

The class is tight, so any sequence has a weakly convergent subsequence. Log-concavity of probability measures is preserved under weak limits [10, Proposition 3.6]. By (2), the limit retains mean zero and variance one. It is therefore nondegenerate and has a log-concave density [10, Theorem 2.7], so lies in $\mathcal{C}$. All fixed absolute moments converge. Convergence also holds in W_q: couple by a common uniform quantile, which gives almost sure convergence, and use the uniform higher moments to obtain uniform integrability of the qth power of the coupling difference. Finally, weak convergence to this nondegenerate log-concave density implies convergence in total variation by [11, Proposition 2]. Its hypothesis that the limit has a density has just been verified; no continuity of that density at its support endpoints is required.

Suppose $\Delta_p(\mu_n) \to 0$. Along any convergent subsequence as above, moment continuity gives $\Delta_p(\mu) = 0$. Theorem 2 then gives $\mu \in \mathcal{A}_p$, while convergence in both distances gives $d_q(\mu_n, \mathcal{A}_p) \to 0$ along this subsequence. A subsequence bounded away from $\mathcal{A}_p$ is consequently impossible. One may therefore take

$$\omega_{p,q}(t) = \sup\{ d_q(\mu, \mathcal{A}_p) : \mu \in \mathcal{C},\ \Delta_p(\mu) \le t \}.$$

This function is finite by (2), and the preceding argument proves its asserted limit (and $\omega_{p,q}(0) = 0$).

For the uniform assertion, the set $\mathcal{A}_p$ is constant on K. If uniformity failed, choose $p_n \in K$ and $\mu_n \in \mathcal{C}$ with $\Delta_{p_n}(\mu_n) \to 0$ but distances bounded away from zero. Pass to $p_n \to p \in K$ and a weakly convergent subsequence of μ_n. The tail bound implies joint continuity of $\mathbb{E}_{\mu_n}|X|^{p_n}$: on bounded intervals the integrands converge uniformly, and the tails are controlled by any fixed moment order larger than $\max K$. The endpoint constants are continuous in p. Hence $\Delta_p(\mu) = 0$, and the same contradiction applies. Taking the supremum also over $p \in K$ defines $\omega_{K,q}$. □

Limits of exponent-uniformity. The fixed-exponent statement includes $p = p^*$, with all three variance-normalised laws in $\mathcal{A}_{p^*}$. It cannot, however, be made uniform through p^* with the exact exponent-dependent equality sets: if $p_n \uparrow p^*$ with $p_n < p^*$, then $\Delta_{p_n}(\nu_+) \to 0$, but $d_{\mathrm{TV}}(\nu_+, \nu_L) > 0$ is fixed. Nor is there such a uniform modulus as $p \to 2$. For the standard Gaussian law γ, $\Delta_p(\gamma) \to 0$ as $p \to 2$, whereas $\min_{\nu \in \{\nu_+, \nu_-\}} d_{\mathrm{TV}}(\gamma, \nu) > 0$. At $p = 2$ every law in $\mathcal{C}$ is an equality law. Thus the corollary concerns convergence to the equality-law set, not selection of a unique branch at the transition, and provides no explicit rate in the moment deficit.

Supplementary material and reproducibility

The bundle contains four supplementary notes with the complete closed forms, derivative identities, enclosure lemmas and box-level details used by the certificates: `supplement/bulk.md` (Propositions 13–14), `supplement/local.md` (Proposition 11), `supplement/strip.md` (Proposition 12) and `supplement/largep.md` (Proposition 6). It also contains the certifiers, their outputs, independent floating-point checks and a one-command replay script (`code/reproduce.sh`). All certificates use python-flint 0.9.0 (Arb) [5] with exact rational region boundaries, outward rounding, and failure on any undecided comparison.

Rigour of the library functions. Arb's elementary and special functions used here (powers, exp, log, Γ, ψ, the upper incomplete gamma function, ${}_1F_1$, ${}_2F_2$ and Tricomi U) return, for ball inputs, a ball containing $f(x)$ for every x in the input balls, or a non-finite ball when this cannot be guaranteed (for instance when an input ball meets a pole or a branch cut). Every certifier treats a non-finite output, like an undecided comparison, as a failure of the box. Integrals in the strip certificate use Arb's rigorous integration [6]: the integrand callback returns enclosures on complex balls and signals non-analyticity (logarithms are evaluated with the analytic flag, so a ball touching the branch cut yields a non-finite value and the box fails). When an external parameter s enters as a ball, every callback value encloses the integrand for all s in that ball, so the resulting enclosure holds uniformly in s. Where monotonicity in a parameter is used for an exponent that may be negative (e.g. $\mathbb{E}(d + sR)^q$ with $q = p - 2 < 0$ in L2), the certifier takes the hull of all corner evaluations, which is valid for either direction of monotonicity.

Proof-to-code map. The domains and exact sign statements are given in the cited propositions; the following table identifies their executable counterparts. Paths in the last two columns are relative to `code/` and `certificates/`. All four supplements are included in the same public archive.

Obligation	Certifier	Output
Bulk B1–B4 (Prop. 14)	`bulk/bulk_cert.py`	`bulk/cert_B*.json`
Crossing (Prop. 13)	`bulk/crossing_cert.py`	`bulk/cert_X.json`
Local L1–L3 (Prop. 11)	`local/local_cert.py`	`local/cert_L*.json`
Strip S1–S2 (Prop. 12)	`strip/strip_cert.py`	`strip/S*_cert.json`
Large p (Prop. 6)	`largep/largep_certify.py`	`largep/largep_cert.json`

The large-p conclusion is a written universal argument; the executable checks its rational constants and optional diagnostics. Numerical samples do not replace any universal claim. In particular the strip helper `G_direct` is a truncated diagnostic, not a certified enclosure of the full function; production S1/S2 use the separately bounded routines.

Status and limitations. This is an unrefereed computer-assisted theorem candidate. Ball arithmetic checks the stated regional inequalities, not the complete written argument in a proof assistant. No unaffiliated reproduction or journal peer review is asserted. The stability corollary is qualitative, and the general sharp two-exponent moment problem remains open here. No arithmetic classification of p^*, including irrationality, is claimed. The public archive is identified by DOI 10.5281/zenodo.23405173.

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